

Booth Multiplication & Unsigned Division

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Signed Multiplication Example

For Signed number,

$$\begin{array}{r}
 11111011 \quad (-5) \\
 \underline{11111101} \quad (-3) \\
 11111011 \\
 0000000X \\
 111011XX \\
 11011XXX \\
 1011XXXX \\
 101XXXXX \\
 10XXXXXX \\
 \underline{1XXXXXXXX} \\
 101 \mid 00001111 \quad (+15)
 \end{array}$$

The "Binary"
Multiplication
Table

*	0	1
0	0	0
1	0	1

We can **calculate it like unsigned numbers** just by **sign extending it** and also by **performing twice many steps** (8 steps vs 4 steps) than unsigned numbers.

Booth Multiplication Algorithm

One of most efficient algorithm to perform multiplication with signed numbers is **Booth Multiplication Algorithm**.

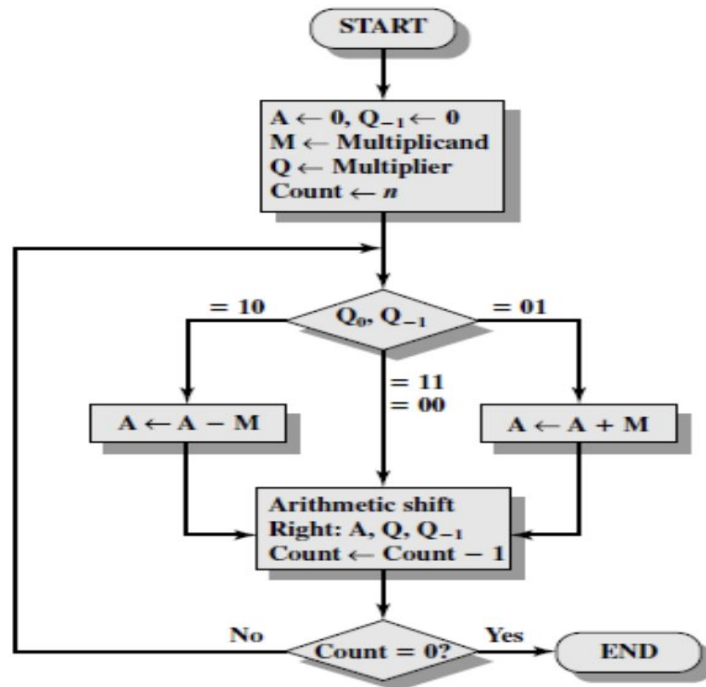


Figure: Booth's Algorithm
for Two's Complement Multiplication

Here,
 $A - M = \text{Signed (2's Complement) Subtraction}$

Booth Multiplication Example

Table: Example of Booth Multiplication (Type 1)

Multiplicand, **M** = 1011 (4-bit)

And Multiplier, **Q** = 1101 (4-bit)

So, Product, **P** = **A,Q** = 00001111 (8-bit)

A	Q	Q ₁	M	Operation	Cycle
0000	110 <u>1</u>	<u>0</u>	1011	Initial Value	
0101 0010	1101 111 <u>0</u>	0 <u>1</u>	1011 1011	A = A-M Shift A,Q to Right	1
1101 1110	1110 111 <u>1</u>	1 <u>0</u>	1011 1011	A = A+M Shift A,Q to Right	2
0011 0001	1111 111 <u>1</u>	0 <u>1</u>	1011 1011	A = A-M Shift A,Q to Right	3
0001 0000	1111 1111	1 1	1011 1011	A = A Shift A,Q to Right	4

Type - 1

Unsigned Division Example

Example: $1111 \div 0111$

Here, Dividend, A = 1111 (15)

Divisor, B = 0111 (7)

If we slightly change its way of calculation, then

0111) 0001111 (0010

- 0000

0011

- 0000

0111

- 0111

0001

- 0000

0001

Quotient = 0010 (2)

Remainder = 0001 (1)

Unsigned Division Example

Table: Example of Unsigned Division

A = 1111 (4-bit) and **B** = 0111 (4-bit)

So, **Q** = 0010 (4-bit) and **R** = 0001(4-bit)

Extend A = 0001011 (7-bit)

Step	A	PA (Partial A)	B	PA-B	Borrow Out	Q	Operation	New PA
1	0001111	<u>0001111</u> <u>0001111</u>	0111	1010	1	0	PA = PA Right Shift PA	<u>0001111</u>
2	0001111	<u>0001111</u> <u>0001111</u>	0111	1100	1	0	PA = PA Right Shift PA	000 <u>1111</u>
3	0001111	<u>0001111</u> <u>0000001</u>	0111	0000	0	1	PA = PA-B Right Shift PA	000 <u>0001</u>
4	0001111	000 <u>0001</u>	0111	1010	1	0	PA = PA	000 <u>0001</u> Remainder